How does a cell receive the right message?

Learning intention

Describe endocrine signalling and explain why suitable internal conditions matter.

A hormone can travel around the body in blood.

Why does every cell not react in the same way?

Transport and conditions

R1. What carries many dissolved substances around the body?

R2. Why do cells need suitable conditions?

R3. Does a chemical messenger think or choose?

LAUNCH SCIENCE I do 4 min

I trace the message route

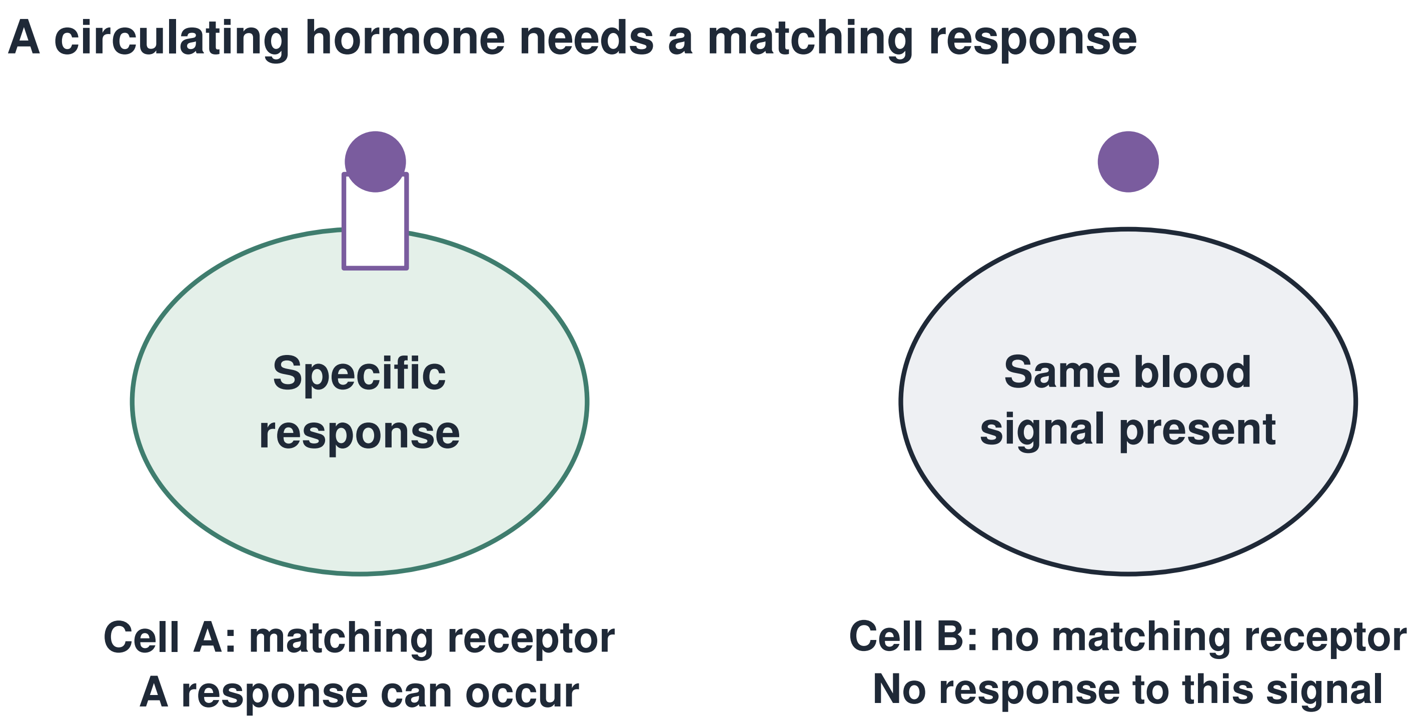
One message, three important locations

Endocrine gland Hormone released into blood	→	Bloodstream Carries hormone	→	Target cells Recognise signal Change activity

The bloodstream distributes the signal.
Cell recognition helps determine the response.

Gland releases → blood carries → target responds. The hormone is a chemical messenger.

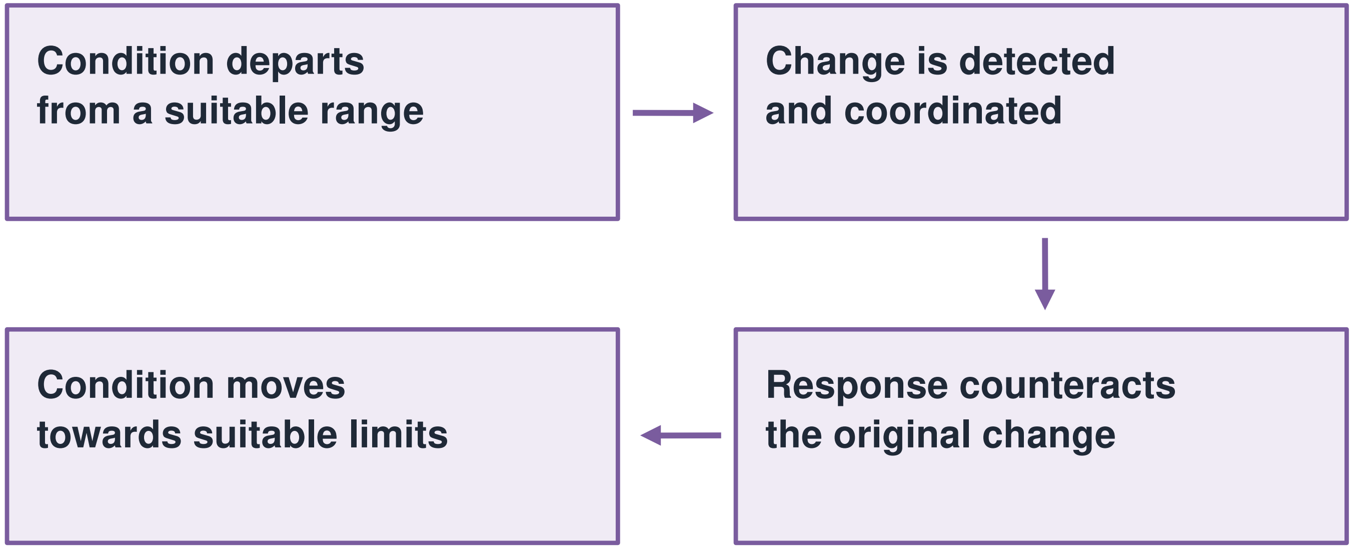
Recognition makes the signal specific



Blood can bring a signal near many cells. A target cell can recognise it and respond.

Homeostasis keeps conditions suitable

Control reduces a departure



Homeostasis is active regulation, not perfect stillness.

Conditions change; regulation limits departures. A corrective response reduces the original change. Some control uses nerves, some uses hormones.

Message-routing desk

W1. Locate the gland names using the position clues. Then trace a pancreas hormone from release to response.

Endocrine structure	Approximate position
Pituitary	Base of brain
Thyroid	Neck
Pancreas	Upper abdomen
Adrenal glands	Above the kidneys
Ovaries	Pelvis
Testes	Scrotum

Schematic location guide; different reproductive organs are listed, not drawn as one anatomy.

Predict the responding cell

A circulating hormone needs a matching response

Cell A: matching receptor
A response can occur

Cell B: no matching receptor
No response to this signal

W2. The same hormone reaches cells A and B. Which cell is more likely to respond in this model?

Control-room repair

W3. “The variable rose, so the controller always pushes it further up.”

Explain what a negative-feedback response should do.

Which explanation links route and recognition?

Choose the mechanism, not a human-like story.

A The hormone circulates; appropriate target cells recognise it and respond.

B The hormone chooses only the cells it likes.

C Homeostasis means no internal value ever changes.

Build a control explanation

Choose your response page.

Use a route diagram and a cause-and-effect sentence.

Make the mechanism visible

Source → transport → target → response.

Explain why suitable conditions matter.

Audience asks about the missing link

L1. Share a short explanation.

Audience: “How did the message reach that cell, and why did it respond?”

A useful model also has a limit

The model shows a mechanism.

It does not predict exact timings, hormone amounts or clinical outcomes.

Three linked ideas

E1. How are endocrine hormones transported?

E2. Why may a non-target cell not respond?

E3. What does homeostasis regulate?

Choose the next useful step

Gland location • signal route • target recognition • control.